

OAS-015 — Reasoning Architecture

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000 through OAS-014
 - OCS-000 through OCS-032
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Purpose

This specification defines the architectural organization of Reasoning.

The Constitution defines the principles of reasoning.

This specification defines how architecture preserves, composes, compares, and evolves reasoning without coupling it to any specific intelligence, algorithm, or implementation.

Reasoning SHALL be architected as a transparent, inspectable, and continuously improvable process.

Architecture SHALL preserve reasoning.

Implementations MAY execute reasoning.

1. Architectural Objective

The objective of Reasoning Architecture is to ensure that every constitutional conclusion remains explainable through the reasoning that produced it.

Architecture SHALL optimize for:

Transparency.

Reconstructability.

Comparability.

Challengeability.

Evolution.

Reasoning SHALL remain visible.

2. Definition

Architectural Reasoning is the organized constitutional process connecting:

Questions

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Evidence

↓

Interpretations

↓

Explanations

↓

Understanding

↓

Decisions

Reasoning SHALL remain independently reconstructable.

3. Separation of Reasoning and Conclusion

Reasoning SHALL remain distinct from conclusions.

Conclusions MAY be preserved.

Reasoning SHALL also be preserved.

Future participants SHALL be capable of evaluating both independently.

4. Multiple Reasoning Paths

Architecture SHALL support multiple concurrent reasoning paths.

Different reasoning paths MAY legitimately produce:

Equivalent conclusions.

Different conclusions.

Complementary conclusions.

Architecture SHALL preserve every path.

5. Reasoning Provenance

Every reasoning process SHALL preserve:

Origin.

Applicable evidence.

Applied assumptions.

Referenced constitutional principles.

Applicable architectural patterns.

Known limitations.

Reasoning provenance SHALL remain explicit.

6. Reasoning Composition

Complex reasoning MAY compose simpler reasoning processes.

Composition SHALL preserve:

Intermediate reasoning.

Dependencies.

Historical continuity.

Explainability.

Architecture SHALL avoid opaque composition.

7. Reasoning Revision

Reasoning MAY evolve.

Revision SHALL preserve:

Previous reasoning.

Reason for revision.

New evidence.

Architectural continuity.

Reasoning SHALL become constitutional history.

8. Human and Artificial Reasoning

Architecture SHALL treat reasoning independently of its source.

Reasoning MAY originate from:

Human Participants.

Artificial Participants.

Collaborative reasoning.

Federated reasoning.

Evaluation SHALL concern the reasoning itself.

Not its origin.

9. Reasoning Verification

Verification SHALL evaluate:

Logical coherence.

Evidence utilization.

Architectural traceability.

Historical reconstruction.

Constitutional compatibility.

Verification SHALL remain implementation-independent.

10. Reasoning Discovery

Architecture SHALL support discovery of reasoning patterns.

Discovery MAY identify:

Recurring assumptions.

Recurring mistakes.

Successful reasoning strategies.

Architectural improvements.

Reasoning SHALL become reusable knowledge.

11. Reasoning Pattern

Architecture SHALL organize reasoning as explicit constitutional pathways rather than hidden computational processes.

Execution MAY differ.

Reasoning SHALL remain observable.

12. Reasoning Architecture Test

Reasoning Architecture satisfies this specification when an independent architect can reconstruct:

Why a conclusion emerged.

Which evidence influenced it.

Which assumptions were applied.

Which alternative reasoning paths existed.

How reasoning evolved.

How future reasoning may improve.

Without requiring implementation-specific computation.

Conformance

An implementation conforms to this specification when reasoning remains transparent, reconstructable, historically attributable, and constitutionally equivalent independently of computational methods or intelligence providers.

Architectural Principle

Conclusions may persuade.

Reasoning teaches.

Architecture therefore preserves not only what was concluded,

but the pathways of thought that made those conclusions possible.

For a discipline that seeks to improve understanding,

the preservation of reasoning is as important as the preservation of knowledge itself.

End of Specification OAS-015